ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Semiconductor

English Medium

EXERCISE-O

SINGLE CORRECT TYPE QUESTIONS

1. An electric field is applied to a semi-conductor. Let the number of charge carriers be n and the average drift speed be v. If the temperature is increased
(A) Both n and v will increase
(B) n will increase but v will decrease
(C) v will increase but n will decrease
(D) Both n and v will decrease

SC0006

2. The manifestation of band structure in solids is due to - **[AIEEE-2004]**
(A) Bohr's correspondence principle
(B) Pauli's exclusion principle
(C) Heisenberg's uncertainty principle
(D) Boltzmann's law

SC0007

3. P-type semiconductor is formed when
A. As impurity is mixed in Si
B. Al impurity is mixed in Si
C. B impurity is mixed in Ge
D. P impurity is mixed in Ge
(A) A and C (B) A and D (C) B and C (D) B and D

SC0008

4. In extrinsic semiconductors -
(A) The conduction band and valence band overlap
(B) The gap between conduction band and valence band is more than 16 eV
(C) The gap between conduction band and valence band is near about 1 eV
(D) The gap between conduction band and valence band will be 100 eV and more

SC0009

5. A strip of copper and another of germanium are cooled from room temperature to 80 K. The resistance of - **[AIEEE-2004]**
(A) copper strip increases and that of germanium decreases
(B) copper strip decreases and that of germanium increases
(C) each of these increases
(D) each of these decreases

SC0010

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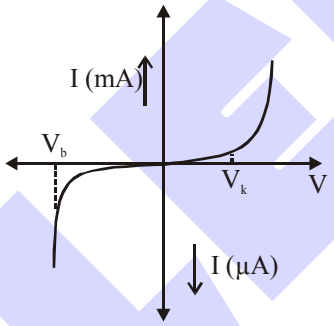
6. In a P-N junction diode not connected to any circuit -
(A) the potential in the same everywhere
(B) the P-type side is at a higher potential then the N-type side
(C) there is an electric field at the junction directed from the N-type side to the P-type side
(D) there is an electric field at the junction directed from the P-type to the N-type side

SC0011

7. When p-n junction diode is forward biased, then. **[AIEEE-2004]**
(A) both the depletion region and barrier height are reduced
(B) the depletion region is widened and barrier height is reduced
(C) the depletion region is reduced and barrier height is increased
(D) both the depletion region and barrier height are increased

SC0012

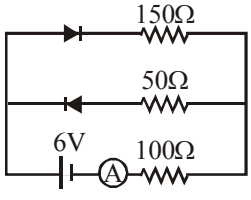
8. The V-I characteristic for a p-n junction diode is plotted as shown in the figure. From the plot we can conclude that
[$V_b \rightarrow$ breakdown voltage, $V_k \rightarrow$ knee voltage]



(A) the forward bias resistance of diode is very high; almost infinity for small values of V and after a certain value it becomes very low
(B) the reverse bias resistance of diode is very high in the beginning upto breakdown voltage is not achieved
(C) both forward and reverse bias resistances are same for all voltages
(D) both (A) and (B) are correct

SC0013

9. In the below given arrangement determine the ammeter reading, if each diodes have a forward resistance of 50 Ω and infinite backward resistance -

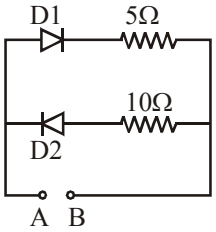


(A) zero (B) 0.02 (C) 0.03 (D) 0.036

SC0014

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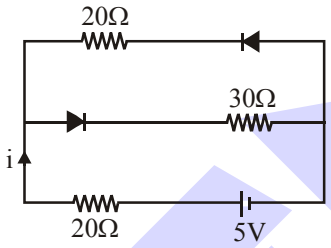
10. A 2V battery is connected across AB as shown in the figure. The value of the current supplied by the battery when in one case battery's positive terminal is connected to A and in other case when positive terminal of battery is connected to B will respectively be :- **(JEE-Main Online-2015)**



(A) 0.1 A and 0.2 A (B) 0.4 A and 0.2 A (C) 0.2 A and 0.4 A (D) 0.2 A and 0.1 A

SC0015

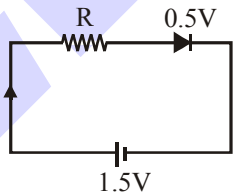
11. Current in the circuit will be -



(A) $\frac{5}{40}$A (B) $\frac{1}{10}$A (C) $\frac{5}{10}$A (D) $\frac{5}{20}$A

SC0016

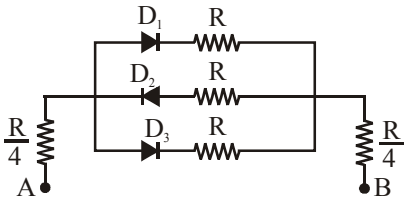
12. The diode used in the circuit shown in the figure has a constant voltage drop of 0.5 V at all currents and a maximum power rating of 100 milliwatts. What should be the value of the resistor R connected in series with the diode for obtaining maximum current -



(A) 1.5 Ω (B) 5 Ω (C) 6.67 Ω (D) 200 Ω

SC0017

13. In the following circuits PN-junction diodes D_1, D_2 and D_3 are ideal for the following potential of A and B, the correct increasing order of resistance between A and B will be -



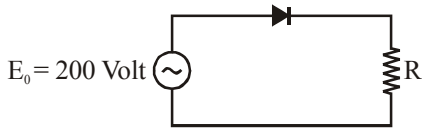
(i) –10V, –5V, (ii) –5V, –10V (iii) –4V, –12V

(A) (i) < (ii) < (iii) (B) (iii) < (ii) < (i) (C) (ii) = (iii) < (i) (D) (i) = (iii) < (ii)

SC0018

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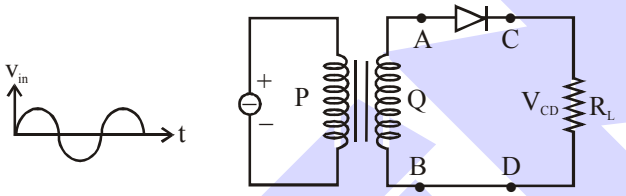
14. A sinusoidal voltage of peak value 200 volt is connected to a diode and resistor R in the circuit shown so that half wave rectification occurs. If the forward resistance of the diode is negligible compared to R then rms voltage (in volt) across R is approximately -

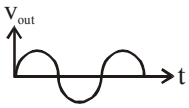


(A) 200 (B) 100 (C) $\frac{200}{\sqrt{2}}$ (D) 280

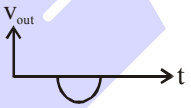
SC0019

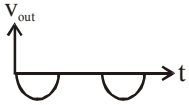
15. In the half-wave rectifier circuit shown. Which one of the following wave forms is true for V_{CD}, if the input is as shown?



(A) 

(B) 

(C) 

(D) 

SC0020

16. Zener diode is a p-n junction which has -
(A) p-end heavily doped, n-end lightly doped
(B) n-end heavily doped, p-end lightly doped
(C) both p and n-ends heavily doped
(D) both p and n-ends lightly doped

SC0021

17. Zener diode has both p and n-ends heavily doped so that -
(A) it has small thickness of depletion region
(B) it has large thickness of depletion region due to large recombination
(C) it has large reverse bias voltage
(D) it has weak reverse current when reverse biased

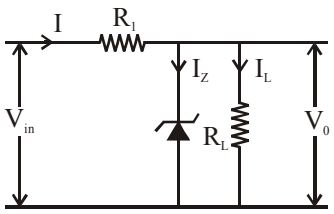
SC0022

18. Most important use of zener diode is to have –
(A) constant voltage across applied load
(B) any desired current at constant voltage
(C) a p-n junction working under constant regulated voltage conditions
(D) a p-n junction to operate at high voltages

SC0023

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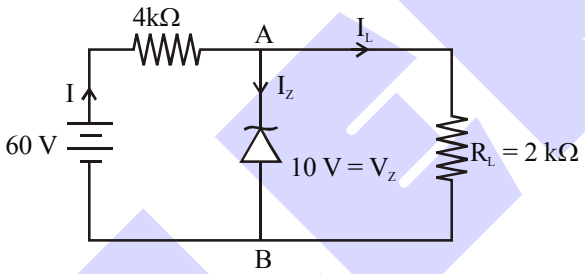
19. In given figure when input voltage increases,



(A) the current through R_S, R_L and zener increases
(B) the current through R_S increases, zener increases but through R_L remains constant
(C) the current through R_S increases, through zener decreases, R_L increases
(D) the current through R_S increases, through zener remains constant but R_L increases

SC0024

20. A Zener diode is connected to a battery and a load as shown below. The currents I, I_Z and I_L are respectively. **(JEE-Main Online-2014)**



(A) 12.5 mA, 5 mA, 7.5 mA
(B) 15 mA, 7.5 mA, 7.5 mA
(C) 12.5 mA, 7.5 mA, 5 mA
(D) 15 mA, 5 mA, 10 mA

SC0025

21. A transistor is used in common emitter mode as an amplifier, then:
(1) the base emitter junction is forward biased
(2) the base emitter junction is reverse biased
(3) the input signal is connected in series with the voltage applied to bias the base emitter junction
(4) the Input signal is connected is series with the voltage applied to bias the base collector junction which is correct-

(A) 1, 2, 3
(B) 1, 2, 3, 4
(C) 1, 3
(D) 2, 3, 4

SC0026

22. In a n-p-n transistor circuit the collector current is 10 mA. If 90% of the electron emitted reach the collector then -
(A) the emitter current will be 9 mA
(B) the base current will be 9 mA
(C) the emitter current will be 11 mA
(D) the base current will be -1 mA

SC0027

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23. n-p-n transistors are preferred to p-n-p transistors because -
(A) they have low cost
(B) they have low dissipation energy
(C) they are capable of handling large power
(D) electrons have high mobility than holes.

SC0028

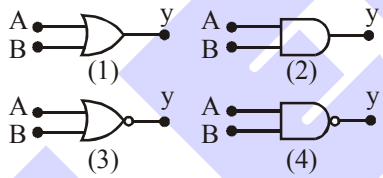
24. When npn transistor is used as an amplifier **[AIEEE-2004]**
(A) electrons move from collector to base
(B) holes move from emitter to base .
(C) electrons move from base to collector
(D) holes move from base to emitter

SC0029

25. For a transistor, α is 0.8. The transistor is in CE configuration. The change in the collector current when the base current changes by 6 mA is :-
(A) 6 mA (B) 4.8mA (C) 24mA (D) 8mA

SC0030

26. Given below are four logic gate symbol (figure). Those for OR, NOR and NAND are respectively



(A) 1,4,3 (B) 4,1,2 (C) 1,3,4 (D) 4,2,1

SC0031

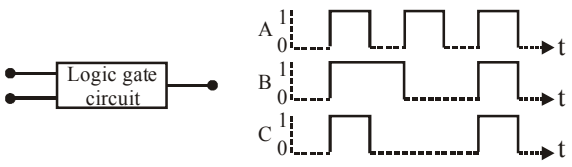
27. The following truth table corresponds to the logic gate -

A	0	0	1	1
B	0	1	0	1
X	0	1	1	1

(A) NAND (B) OR (C) AND (D) XOR

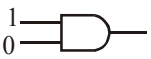
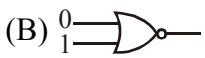
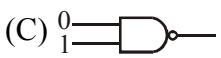
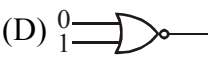
SC0032

28. The following figure shows a logic gate circuit with two input A and B output C. The voltage waveforms of A, B and C are as shown in second figure below. The logic gate is :



(A) OR gate (B) NAND gate (C) AND gate (D) NOR gate

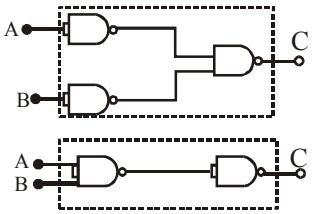
SC0033

29. Which of the following gates will have an output of 1 -
(A)  (B)  (C)  (D) 

SC0034

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30. The combination of 'NAND' gates shown here under (figure) are equivalent to -

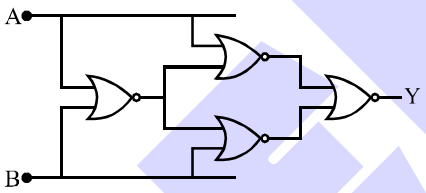


(A) An OR gate and an AND gate respectively
(B) An AND gate and a NOT gate respectively
(C) An AND gate and an OR gate respectively
(D) An OR gate and a NOT gate respectively

SC0035

31. A system of four gates is set up as shown.The 'truth table' corresponding to this system is :-

(JEE-Main Online-2013)



(A)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(B)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(C)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(D)

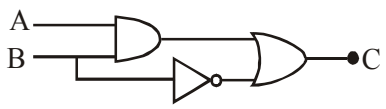
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

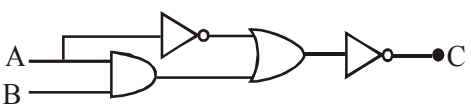
SC0036

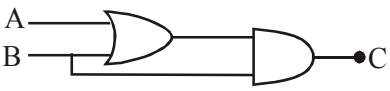
32. Which of the following circuits correctly represents the following truth table ?

(JEE-Main Online-2013)

A	B	C
0	0	0
0	1	0
1	0	1
1	1	0

(A) 

(B) 

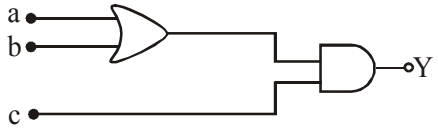
(C) 

(D) 

SC0037

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33. To get an output of 1 from the circuit shown in figure the input must be :- **(JEE-Main Online-2016)**

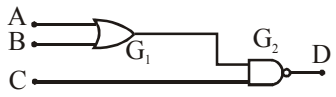


(A) a = 1, b = 0, c = 1 (B) a = 1, b = 0, c = 0

(C) a = 0, b = 1, c = 0 (D) a = 0, b = 0, c = 1

SC0038

34. For the given combination of gates, if the logic states of inputs A, B, C are as follows A = B = C = 0 and A = B = 1, C = 0 then the logic states of output D are -



(A) 0, 0 (B) 0, 1 (C) 1, 0 (D) 1,1

SC0039

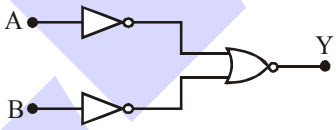
35. This symbol represents



(A) NOT gate (B) OR (C) AND gate (D) NOR gate

SC0040

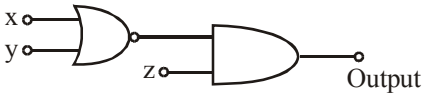
36. Which logic gate is represented by the following combination of logic gates -



(A) OR (B) NAND (C) AND (D) NOR

SC0041

37. In which of the following cases, we would obtain an output of one.



(A) x = l, y = l, z = l (B) x = l, y = l, z = 0

(C) x = 0, y = l, z = l (D) x = 0, y = 0, z = l

SC0042

38. In a full wave rectifier circuit operating from 50 Hz mains frequency, the fundamental frequency in the ripple would be :- **[AIEEE-2005]**

(A) 50 Hz (B) 25Hz (C) 100 Hz (D) 70.7 Hz

SC0043

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39. The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than 2480 nm is incident on it. The band gap (in eV) for the semiconductor is :- **[AIEEE-2005]**

(A) 1.1 eV (B) 2.5 eV (C) 0.5 eV (D) 0.7 eV

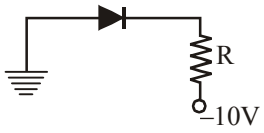
SC0044

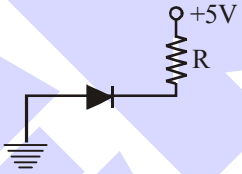
40. If the ratio of the concentration of electrons to that of holes in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$ then what is the ratio of their drift velocities - **[AIEEE-2006]**

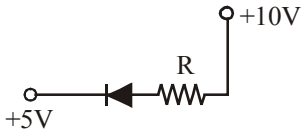
(A) $\frac{5}{4}$ (B) $\frac{4}{7}$ (C) $\frac{5}{6}$ (D) $\frac{4}{5}$

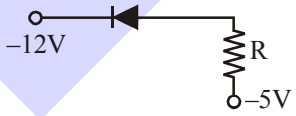
SC0045

41. In the following, which one of the diodes is reverse biased ; **[AIEEE-2006]**

(A) 

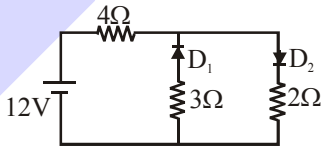
(B) 

(C) 

(D) 

SC0046

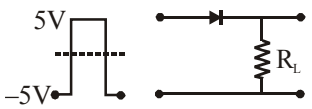
42. The circuit has two oppositely connected ideal diodes in parallel. What is the current flowing in the circuit ? **[AIEEE-2006]**

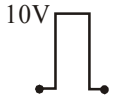


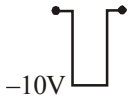
(A) 2.31 A (B) 1.33 A (C) 1.71 A (D) 2.00 A

SC0047

43. If in a p-n junction diode, a square input signal of 10V is applied as shown. Then the output signal across R_L will be :- **[AIEEE-2007]**



(A) 

(B) 

(C) 

(D) 

SC0048

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2, EM & INS. Semiconductor-PCS\Eng\4-Sanic\01_Th + Exp\65

44. Carbon, silicon and germanium have four valence electrons each. At room temperature which one of the following statements is most appropriate ? **[AIEEE-2007]**

(A) The number of free conduction electrons is significant in C but small in Si and Ge

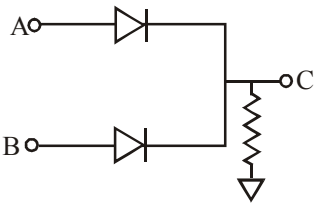
(B) The number of free conduction electrons is negligibly small in all the three

(C) The number of free electrons for conduction is significant in all the three

(D) The number of free electrons for conduction is significant only in Si and Ge but small in C

SC0049

45. In the circuit below, A and B represent two inputs and C represents the output. **[AIEEE-2008]**

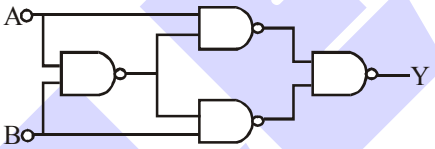


The circuit represents

(A) AND gate (B) NAND gate (C) OR gate (D) NOR gate

SC0055

46. Truth table for system of four NAND gates as shown in figure is: **[AIEEE-2012]**



(A)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(B)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(C)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(D)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

SC0059

MULTIPLE CORRECT TYPE QUESTIONS

47. Which of the following statements is correct

(A) Resistance of semiconductor decreases with increase in temperature

(B) In an electric field, displacement of holes is opposite to the displacement of electrons

(C) Resistance of a conductor decreases with the increase in temperature

(D) n-type semiconductors are neutral

SC0050

48. Pick out the correct statement the reverse current in p-n junction diode

(A) can be minimum and constant

(B) remains constant even after the breakdown voltage

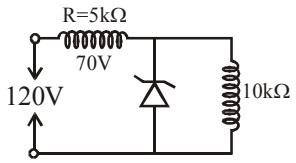
(C) becomes infinity at breakdown

(D) reverse current is controlled by external resistance

SC0051

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM&INS.-Semiconductor-PCS\Eng\V4-Semi\01_Th + Exp\65

49. For the circuit shown in the figure:



(A) current through zener diode is 4 mA
(B) current through zener diode is 9 mA
(C) the output voltage is 50 V
(D) the output voltage is 40 V

SC0052

50. Which of the following devices are heavily doped p-n junction
(A) Photo diode
(B) Light emitting diode
(C) Solar cell
(D) Zener mode

SC0053

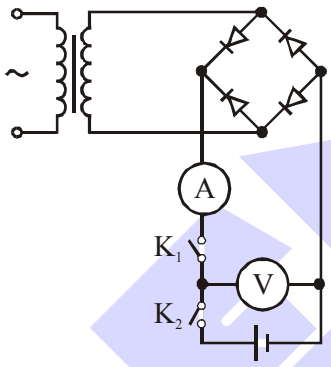
51. Which of the following statements is correct for proper working of zener diode ?
(A) Reverse bias voltage should be less than or equal to zener breakdown voltage
(B) Reverse bias voltage applied must be greater than zener breakdown voltage.
(C) Zener is to be reverse biased for zener action
(D) For given zener diode there can be different zener breakdown voltages

SC0054

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2 EM & INS. Semiconductor-PCS\Eng\4-Sami\01_Th + Exp65

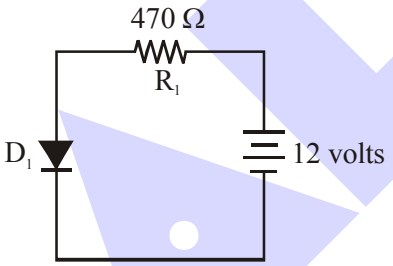
EXERCISE-S

1. The battery is charged from full wave rectifier fed by a sinusoidal voltage (see figure). Ideal diodes, ammeter and voltmeter show the time-average value. At idle with only key K_1 closed, voltmeter shows 12 V, and current is then absent, ie, reading of ammeter is 0. If only the key K_2 is closed, the voltmeter shows battery voltage at 12.3 V. During charging, when the K_2 and K_1 are closed, the voltmeter shows 12.8 V and ammeter shows 5 A. Find the internal resistance of battery.



SC0001

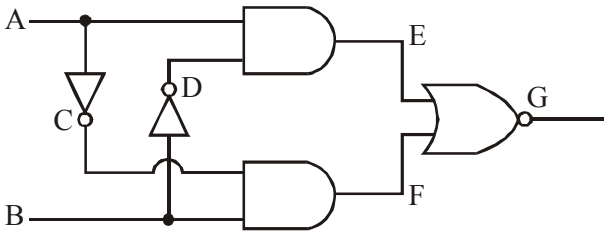
2. Complete the following table of values for this diode circuit, assuming a typical forward voltage drop of 0.65 volts for the diode:



	R_1	D_1	Total
V			12 V
I			
R	470 Ω		
P			

SC0002

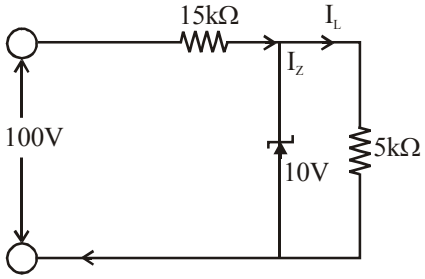
3. Write a truth table for the circuit in figure, including the states at C, D, E, F and G.



SC0003

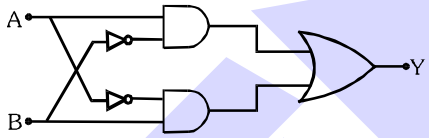
node06\B0BC-BD\Kota\IEE\Advanced\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM&INS.-Semiconductor-PCS\Eng\V4-Semi\01_Th + Exp\65

4. For the circuit shown in figure, find
i) the output voltage
ii) the voltage drop across series resistance
iii) the current through zener diode.



SC0004

5. Write down the actual logic operation carried by the following circuit. Explain your answer.



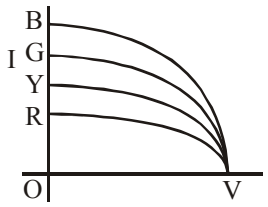
SC0005

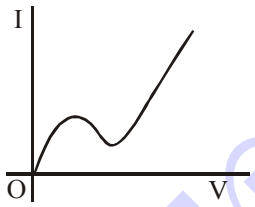
ALLEN®

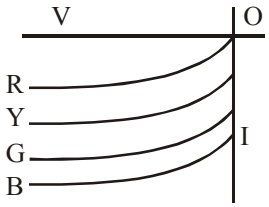
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2 EM & INS. Semiconductor-PCS\Eng\4-Sami\01_Th + Exp65

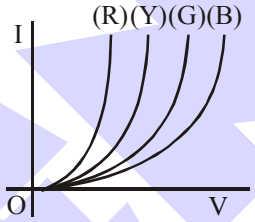
EXERCISE-JM

1. The I-V characteristic of an LED is **[JEE Main 2013]**

(1) 

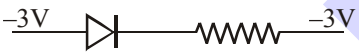
(2) 

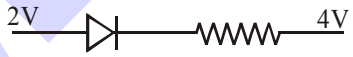
(3) 

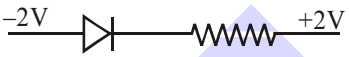
(4) 

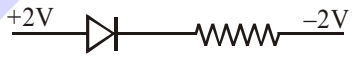
SC0060

2. The forward biased diode connection is: **[JEE Main-2014]**

(1) 

(2) 

(3) 

(4) 

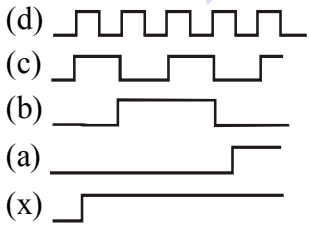
SC0061

3. For a common emitter configuration, if α and β have their usual meanings, the **incorrect** relationship between α and β is **[JEE Main-2016]**

(1) $\alpha = \frac{\beta^2}{1+\beta^2}$ (2) $\frac{1}{\alpha} = \frac{1}{\beta} + 1$ (3) $\alpha = \frac{\beta}{1-\beta}$ (4) $\alpha = \frac{\beta}{1+\beta}$

SC0062

4. If a, b, c, d are inputs to a gate and x is its output, then as per the following time graph, the gate is **[JEE Main-2016]**



(1) NAND (2) NOT (3) AND (4) OR

SC0063

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM&INS.-Semiconductor-PCS\Eng\Vol-Semi\01_Th + Exp\65

5. The temperature dependence of resistances of Cu and undoped Si in the temperature range 300-400K, is best described by :- **[JEE Main-2016]**

(1) Linear decrease for Cu, linear decrease for Si.

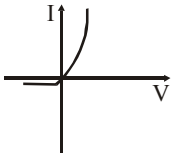
(2) Linear increase for Cu, linear increase for Si.

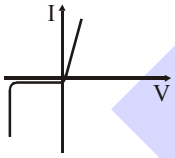
(3) Linear increase for Cu, exponential increase for Si

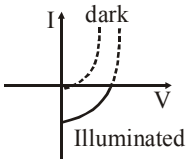
(4) Linear increase for Cu, exponential decrease for Si

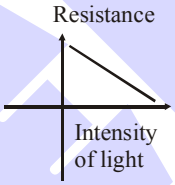
SC0064

6. Identify the semiconductor devices whose characteristics are given below, in the order (a), (b), (c), (d):- **[JEE Main-2016]**

(a) 

(b) 

(c) 

(d) 

(1) Zener diode, Solar cell, Simple diode, Light dependent resistance

(2) Simple diode, Zener diode, Solar cell, Light dependent resistance

(3) Zener diode, Simple diode, Light dependent resistance, Solar cell

(4) Solar cell, Light dependent resistance, Zener diode, Simple diode

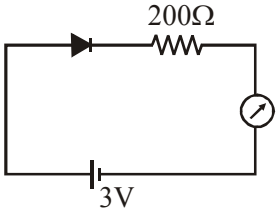
SC0065

7. In a common emitter amplifier circuit using an n-p-n transistor, the phase difference between the input and the output voltages will be : **[JEE Main-2017]**

(1) 135° (2) 180° (3) 45° (4) 90°

SC0066

8. The reading of the ammeter for a silicon diode in the given circuit is :- **[JEE Main-2018]**



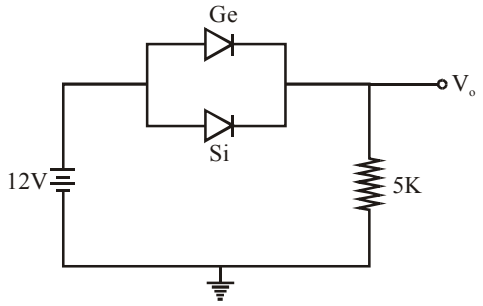
(1) 15 mA (2) 11.5 mA (3) 13.5 mA (4) 0

SC0067

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2 EM & INS. Semiconductor-PCS\Eng\4-Sami\01_Th + Exp\65

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

9. Ge and Si diodes start conducting at 0.3 V and 0.7 V respectively. In the following figure if Ge diode connection are reversed, the value of V_o changes by : (assume that the Ge diode has large breakdown voltage) **[JEE Main-2019_Jan]**



(1) 0.6 V (2) 0.8 V (3) 0.4 V (4) 0.2 V

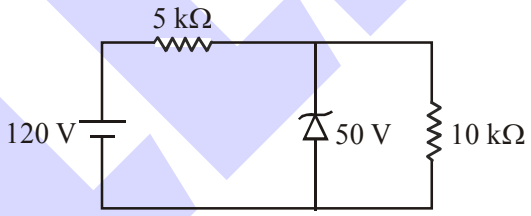
SC0068

10. Mobility of electrons in a semiconductor is defined as the ratio of their drift velocity to the applied electric field. If, for an n-type semiconductor, the density of electrons is $10^{19}m^{-3}$ and their mobility is 1.6 $m^2/(V.s)$ then the resistivity of the semiconductor (since it is an n-type semiconductor contribution of holes is ignored) is close to: **[JEE Main-2019_Jan]**

(1) 2Ωm (2) 0.4Ωm (3) 4Ωm (4) 0.2Ωm

SC0069

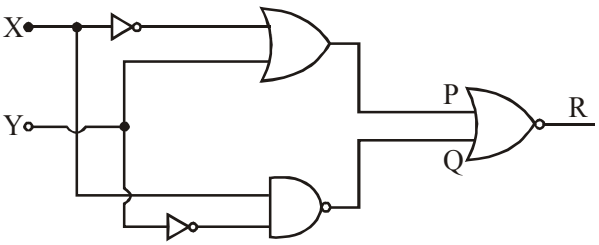
11. For the circuit shown below, the current through the Zener diode is : **[JEE Main-2019_Jan]**



(1) 5 mA (2) Zero (3) 14 mA (4) 9 mA

SC0070

12. To get output '1' at R, for the given logic gate circuit the input values must be : **[JEE Main-2019_Jan]**



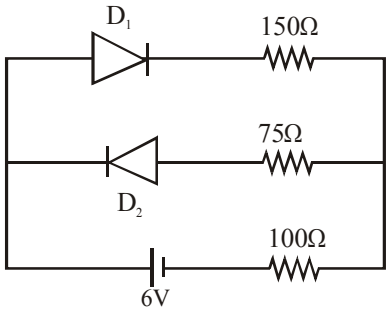
(1) X = 0, Y = 1 (2) X = 1 , Y = 1 (3) X = 0, Y = 0 (4) X = 1, Y = 0

SC0071

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM&INS.-Semiconductor-PCS\Eng\v4-Semi\01_Th + Exp\65

13. The circuit shown below contains two ideal diodes, each with a forward resistance of 50Ω. If the battery voltage is 6 V, the current through the 100 Ω resistance (in Amperes) is :-

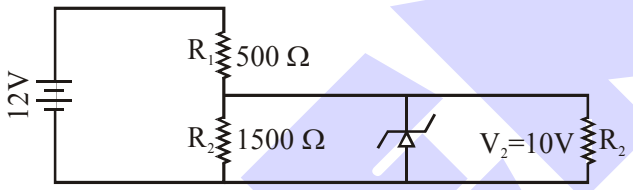
[JEE Main-2019_Jan]



(1) 0.027 (2) 0.020 (3) 0.030 (4) 0.036

SC0072

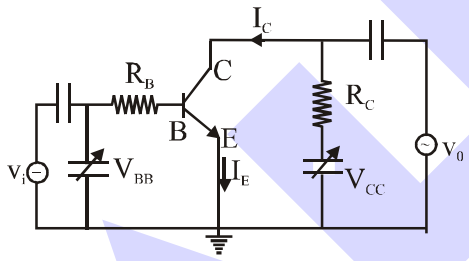
14. In the given circuit the current through Zener Diode is close to : **[JEE Main-2019_Jan]**



(1) 6.0 mA (2) 4.0 mA (3) 6.7 mA (4) 0.0 mA

SC0073

15.

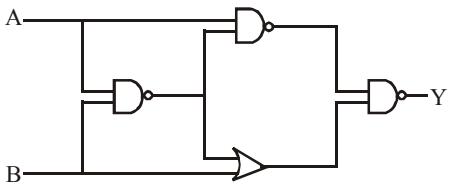


In the figure, given that V_{BB} supply can vary from 0 to 5.0 V, V_{CC} = 5V, β_{dc} = 200, R_B = 100 kΩ, R_C = 1 kΩ and V_{BE} = 1.0 V. The minimum base current and the input voltage at which the transistor will go to saturation, will be, respectively : **[JEE Main-2019_Jan]**

(1) 20μA and 3.5V (2) 25μA and 3.5V (3) 25μA and 2.8V (4) 20μA and 2.8V

SC0074

16. The output of the given logic circuit is : **[JEE Main-2019_Jan]**

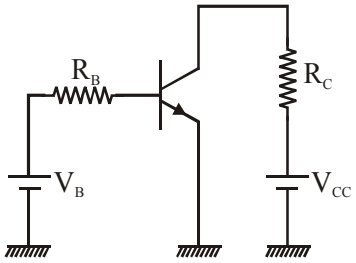


(1) $\bar{A}B$ (2) $A\bar{B}$ (3) $AB+\overline{AB}$ (4) $A\bar{B}+\bar{A}B$

SC0075

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2, EM & INS. Semiconductor-PCS\Eng\4-Sanic\01_Th + Exp65

17. A common emitter amplifier circuit, built using an npn transistor, is shown in the figure. Its dc current gain is 250, R_C = 1kΩ and V_{CC} = 10 V. What is the minimum base current for V_{CE} to reach saturation ?
[JEE Main-2019_April]



(1) 100 µA (2) 7 µA (3) 40 µA (4) 10 µA

SC0076

18. An NPN transistor is used in common emitter configuration as an amplifier with 1 kΩ load resistance. Signal voltage of 10 mV is applied across the base-emitter. This produces a 3 mA change in the collector current and 15µA change in the base current of the amplifier. The input resistance and voltage gain are :
[JEE Main-2019_April]

(1) 0.33 kΩ, 1.5 (2) 0.67 kΩ, 200 (3) 0.33 kΩ, 300 (4) 0.67 kΩ, 300

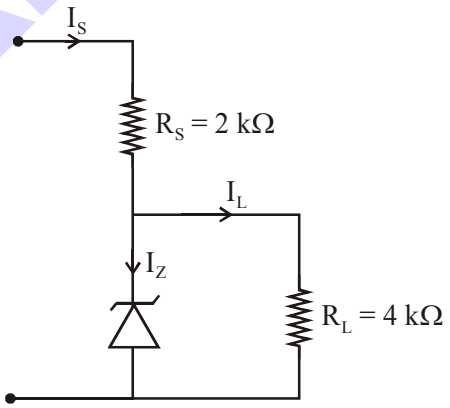
SC0077

19. An npn transistor operates as a common emitter amplifier, with a power gain of 60 dB. The input circuit resistance is 100Ω and the output load resistance is 10 kΩ. The common emitter current gain β is :
[JEE Main-2019_April]

(1) 60 (2) 10^4 (3) 6×10^2 (4) 10^2

SC0078

20. Figure shown a DC voltage regulator circuit, with a Zener diode of breakdown voltage = 6V. If the unregulated input voltage varies between 10 V to 16 V, then what is the maximum Zener current ?
[JEE Main-2019_April]

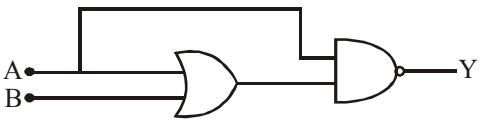


(1) 2.5 mA (2) 3.5 mA (3) 7.5 mA (4) 1.5 mA

SC0079

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM&INS.-Semiconductor-PCS\Eng\04-Semi\01_Th + Exp\65

21. The truth table for the circuit given in the fig. is: **[JEE Main-2019_April]**



(1)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(2)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(3)

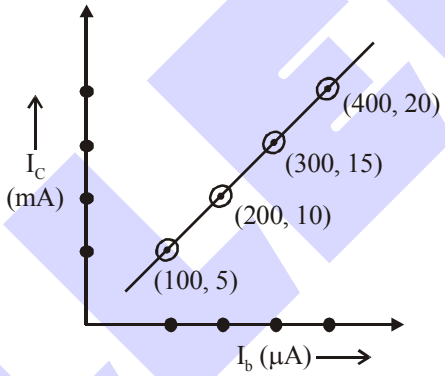
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(4)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

SC0080

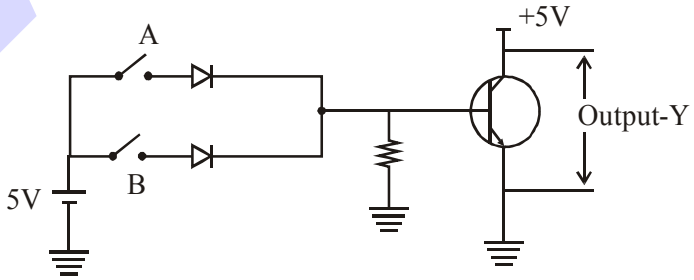
22. The transfer characteristic curve of a transistor, having input and output resistance 100 Ω and 100 kΩ respectively, is shown in the figure. The Voltage and Power gain, are respectively : **[JEE Main-2019_April]**



(1) 5×10^4, 5×10^5
(2) 5×10^4, 5×10^6
(3) 5×10^4, 2.5×10^6
(4) 2.5×10^4, 2.5×10^6

SC0081

23. Boolean relation at the output stage-Y for the following circuit is : **[JEE Main-2020_Jan]**

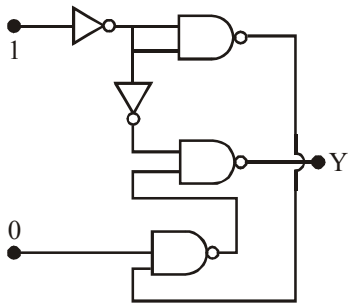


(1) A + B
(2) $\overline{A} + \overline{B}$
(3) $\overline{A} \cdot \overline{B}$
(4) $A \cdot B$

SC0082

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2 EM & INS. Semiconductor-PCS\Eng\4-Sami\01_Th + Exp65

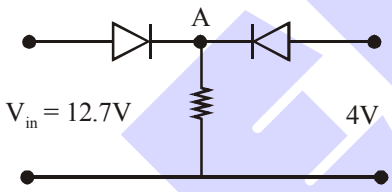
24. In the given circuit, value of Y is : **[JEE Main-2020_Jan]**



(1) will not execute (2) 0
(3) toggles between 0 and 1 (4) 1

SC0083

25. Both the diodes used in the circuit shown are assumed to be ideal and have negligible resistance when these are forward biased. Built in potential in each diode is 0.7 V. For the input voltages shown in the figure, the voltage (in Volts) at point A is _____ . **[JEE Main-2020_Jan]**



SC0084

26. If a semiconductor photodiode can detect a photon with a maximum wavelength of 400 nm, then its band gap energy is: **[JEE Main-2020_SEP]**

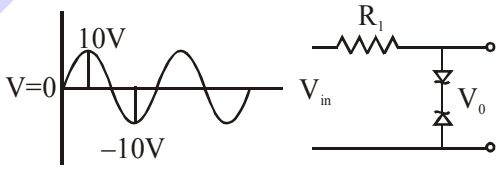
Planck's constant $h = 6.63 \times 10^{-34}$ J.s.

Speed of ligh $c = 3 \times 10^{8}$ m/s

(1) 2.0 eV (2) 1.5 eV (3) 3.1 eV (4) 1.1 eV

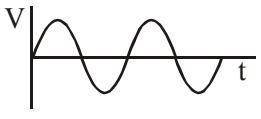
SC0085

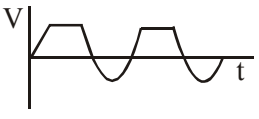
27. Take the breakdown voltage of the zener diode used in the given circuit as 6V. For the input voltage shown in figure below, the time variation of the output voltage is : (Graphs drawn are schematic and not to scale) **[JEE Main-2020_SEP]**



(1) 

(2) 

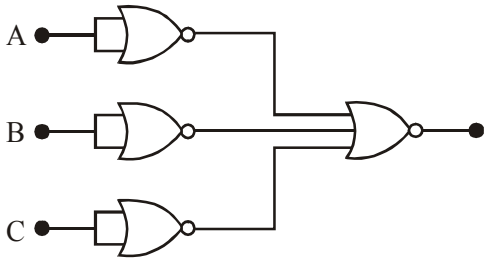
(3) 

(4) 

SC0086

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM&INS.-Semiconductor-PCS\Eng\V4-Semi\01_Th + Exp\65

28. Identify the operation performed by the circuit given below : **[JEE Main-2020_SEP]**



(1) AND (2) NAND (3) OR (4) NOT

SC0087

29. With increasing biasing voltage of a photodiode, the photocurrent magnitude :**[JEE Main-2020_SEP]**

(1) increases initially and saturates finally

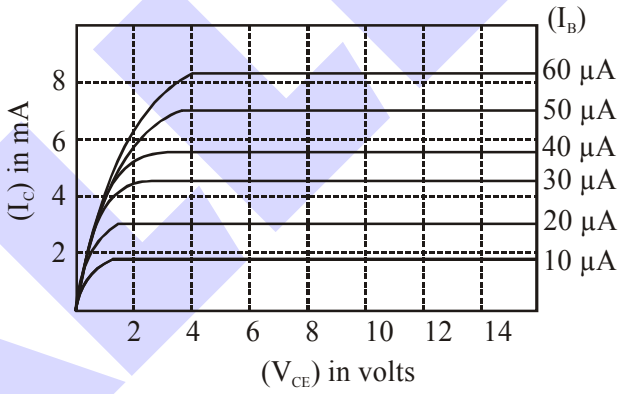
(2) increases initially and after attaining certain value, it decreases

(3) increases linearly

(4) remains constant

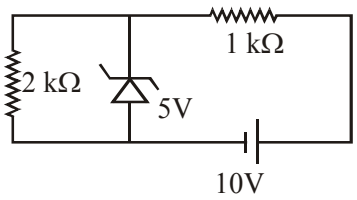
SC0088

30. The output characteristics of a transistor is shown in the figure. When V_{CE} is 10 V and 1_C = 4.0 mA, then value of β_{ac} is _____ . **[JEE Main-2020_SEP]**



SC0089

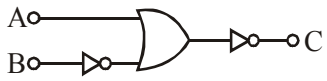
31. In connection with the circuit drawn below, the value of current flowing through 2 kΩ resistor is _____ $\times 10^{-4}$ A. **[JEE Main-2021_Feb]**



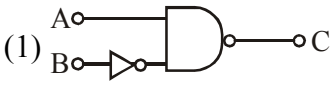
SC0090

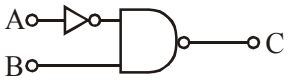
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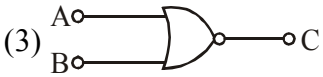
32.

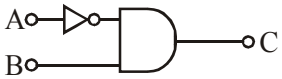


The logic circuit shown above is equivalent to : **[JEE Main-2021_Feb]**

(1) 

(2) 

(3) 

(4) 

SC0091

33. Zener breakdown occurs in a *p-n* junction having p and n both : **[JEE Main-2021_Feb]**

(1) lightly doped and have wide depletion layer.
(2) heavily doped and have narrow depletion layer.
(3) lightly doped and have narrow depletion layer.
(4) heavily doped and have wide depletion layer.

SC0092

34. For extrinsic semiconductors; when doping level is increased; **[JEE Main-2021_Feb]**

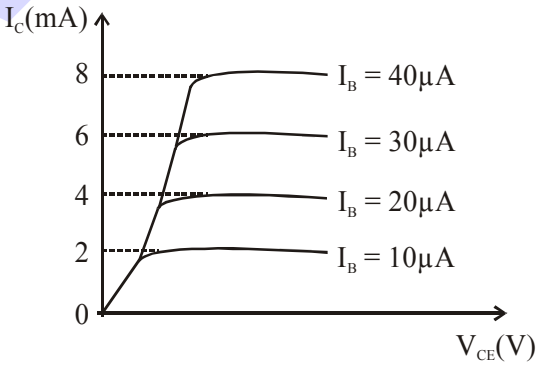
(1) Fermi-level of p-type semiconductor will go upward and Fermi-level of n-type semiconductors will go downward.
(2) Fermi-level of p-type semiconductors will go downward and Fermi-level of n-type semiconductor will go upward.
(3) Fermi-level of both p-type and n-type semiconductros will go upward for $T > T_F$ K and downward for $T < T_F$ K, where T_F is Fermi temperature.
(4) Fermi-level of p and n-type semiconductors will not be affected.

SC0093

35. An npn transistor operates as a common emitter amplifier with a power gain of 10^6. The input circuit resistance is 100Ω and the output load resistance is 10 KΩ. The common emitter current gain 'β' will be _____. (Round off to the Nearest Integer) **[JEE Main-2021_March]**

SC0095

36. The typical output characteristics curve for a transistor working in the common-emitter configuration is shown in the figure. **[JEE Main-2021_March]**

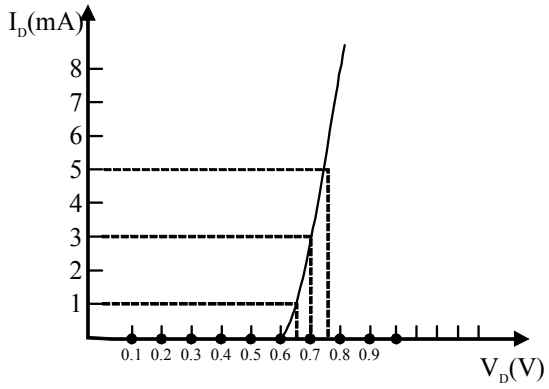


The estimated current gain from the figure is

SC0096

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM&INS.-Semiconductor-PCS\Eng\V4-Semi\01_Th + Exp\65

37. For the forward biased diode characteristics shown in the figure, the dynamic resistance at $I_D = 3$ mA will be_____Ω. **[JEE Main-2021_July]**

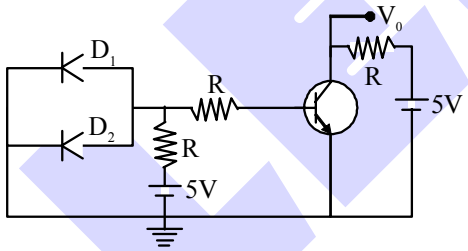


SC0097

38. In a semiconductor, the number density of intrinsic charge carriers at 27°C is $1.5 \times 10^{16}/m^3$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5 \times 10^{22}/m^3$. The electron density in the doped semiconductor is _____ $\times 10^9/m^3$. **[JEE Main-2021_July]**

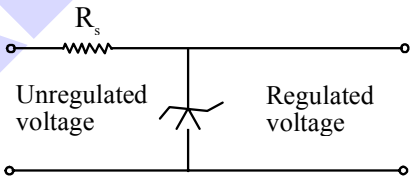
SC0098

39. A circuit is arranged as shown in figure. The output voltage V_0 is equal to ..... V. **[JEE Main-2021_Aug]**



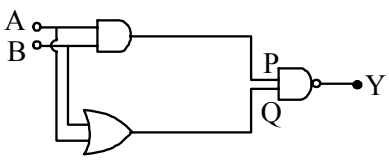
SC0099

40. A zener diode of power rating 2W is to be used as a voltage regulator. If the zener diode has a breakdown of 10 V and it has to regulate voltage fluctuated between 6 V and 14 V, the value of R_s for safe operation should be _____ Ω. **[JEE Main-2021_Aug]**



SC0100

41. In the following logic circuit the sequence of the inputs A, B are (0, 0), (0,1), (1, 0) and (1, 1). The output Y for this sequence will be : **[JEE Main-2021_Aug]**

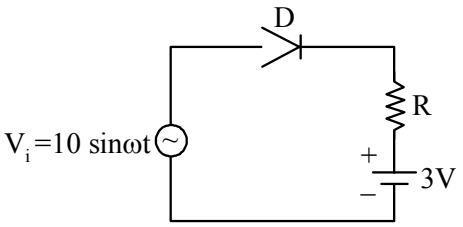


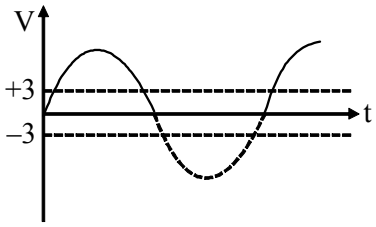
(1) 1, 0, 1, 0 (2) 0, 1, 0, 1 (3) 1, 1, 1, 0 (4) 0, 0, 1, 1

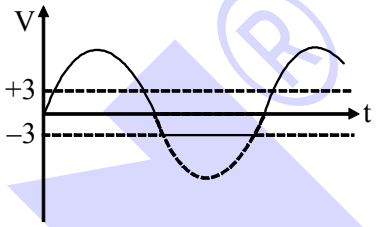
SC0101

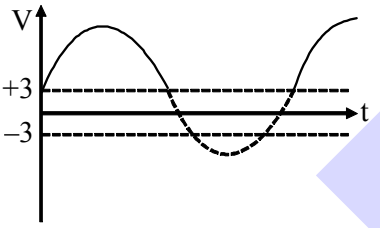
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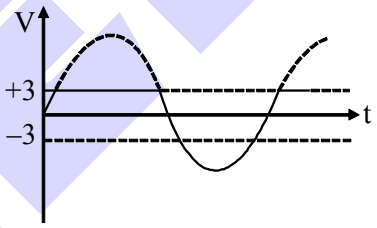
42. Choose the correct waveform that can represent the voltage across R of the following circuit, assuming the diode is ideal one: **[JEE Main-2021_Aug]**



(1)


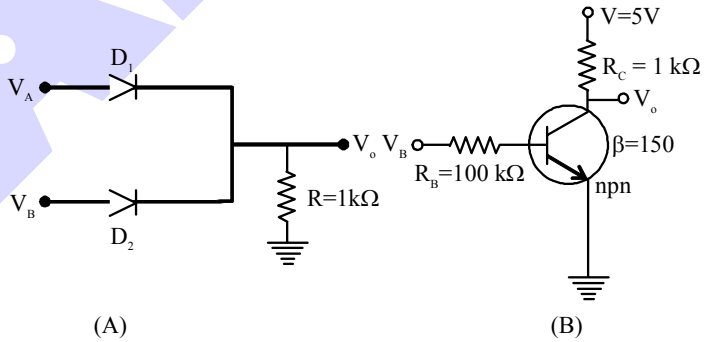
(2)


(3)


(4)


SC0102

43. If V_A and V_B are the input voltages (either 5V or 0V) and V_0 is the output voltage then the two gates represented in the following circuit (A) and (B) are:- **[JEE Main-2021_Aug]**



(1) AND and OR Gate
(2) OR and NOT Gate
(3) NAND and NOR Gate
(4) AND and NOT Gate

SC0103

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM&INS.-Semiconductor-PCS\Eng\V4-Semi\01_Th + Exp\65

44. **Statement-I :**

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect a capacitor across the output parallel to the load R_L. **[JEE Main-2021_Aug]**

Statement-II :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect an inductor in series with R_L.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) **Statement I** is true but **Statement II** is false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are false

(4) Both **Statement I** and **Statement II** are true

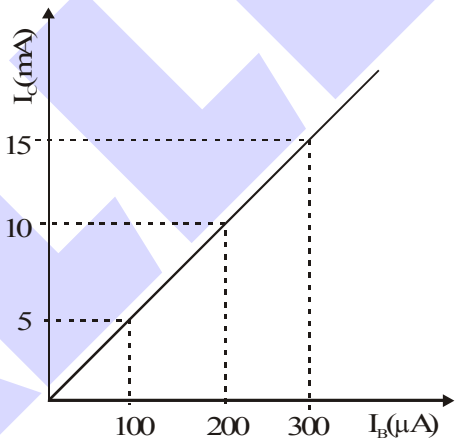
SC0104

45. A transistor is used in common-emitter mode in an amplifier circuit. When a signal of 10 mV is added to the base-emitter voltage, the base current changes by 10 μA and the collector current changes by 1.5 mA. The load resistance is 5 kΩ. The voltage gain of the transistor will be _____.

[JEE Main-2022_June]

SC0105

46. In an experiment of CE configuration of n-p-n transistor, the transfer characteristics are observed as given in figure.



If the input resistance is 200Ω and output resistance is 60Ω the voltage gain in this experiment will be _____

[JEE Main-2022_June]

SC0106

47. The photodiode is used to detect the optiocal signals. These diodes are preferably operated in reverse biased mode because. **[JEE Main-2022_June]**

(A) fractional change in majority carriers produce higher forward bias current

(B) fractional change in majority carriers produce higher reverse bias current

(C) fractional change in minority carriers produce higher forward bias current

(D) fractional change in minority carriers produce higher reverse bias current

SC0107

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2 EM & INS. Semiconductor-PCS\Eng\4-Sami\01_Th + Exp65

48. The positive feedback is required by an amplifier to act an oscillator. The feedback here means :

[JEE Main-2022_June]

(A) External input is necessary to sustain ac signal in output.

(B) A portion of the output power is returned back to the input.

(C) Feedback can be achieved by LR network.

(D) The base-collector junction must be forward biased. **SC0108**

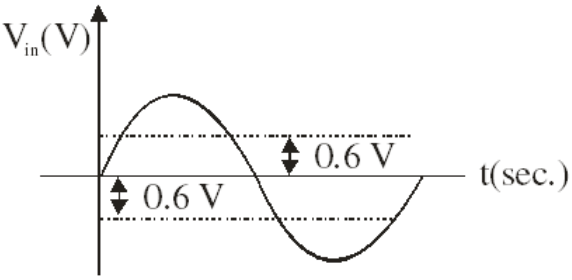
49. For a transistor to act as a switch, it must be operated in **[JEE Main-2022_June]**

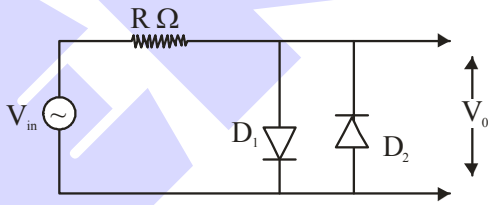
(A) Active region (B) Saturation state only

(C) Cut-off state only (D) Saturation and cut-off state **SC0109**

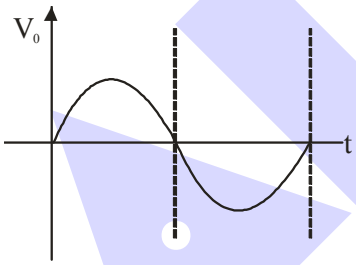
50. In the given circuit the input voltage V_{in} is shown in figure. The cut-in voltage of p-n junction diode (D_1 or D_2) is 0.6 V. Which of the following output voltage (V_0) waveform across the diode is correct?

[JEE Main-2022_June]

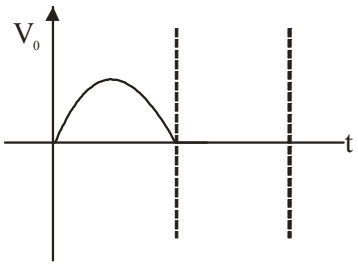




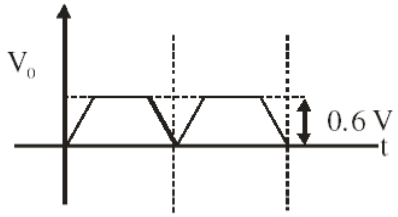
(A)



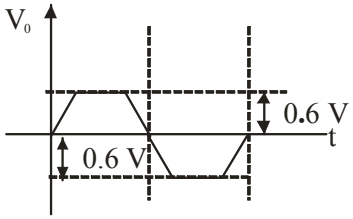
(B)



(C)



(D)

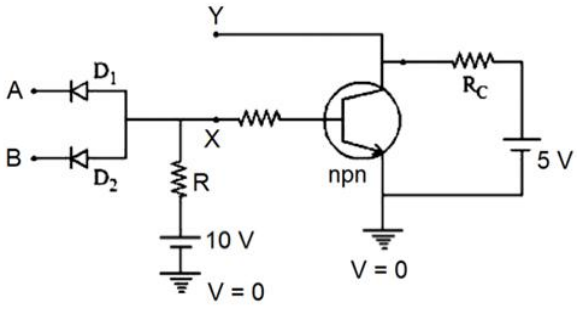


SC0110

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51. In the following circuit, the correct relation between output (Y) and inputs A and B will be :

[JEE Main-2022_June]



(A) $Y = AB$ (B) $Y = A + B$ (C) $Y = \overline{AB}$ (D) $Y = \overline{A + B}$

SC0111

52. For using a multimeter to identify diode from electrical components. choose the correct statement out of the following about the diode : **[JEE Main-2022_June]**

(A) It is two terminal device which conducts current in both directions.

(B) It is two terminal device which conducts current in one direction only

(C) It does not conduct current gives an initial deflection which decays to zero.

(D) It is three terminal device which conducts current in ne direction only between central terminal and either of the remaining two terminals

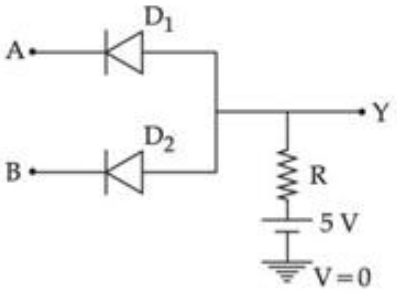
SC0112

53. A potential barrier of 0.4 V exists across a p-n junction. An electron enters the junction from the n-side with a speed of 6.0×10^5 ms^{-1}. The speed with which electron enters the p side will be $\frac{x}{3} \times 10^5 ms^{-1}$ the value of x is _____. **[JEE Main-2022_June]**

(Given mass of electron = 9×10^{-31} kg, charge on electron = 1.6×10^{-19}C.)

SC0113

54. In the circuit, the logical value of A = 1 or B = 1 when potential at A or B is 5V and the logical value of A = 0 or B = 0 when potential at A or B is 0 V.



The truth table of the given circuit will be : **[JEE Main-2022_July]**

(A)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

(B)

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

(C)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

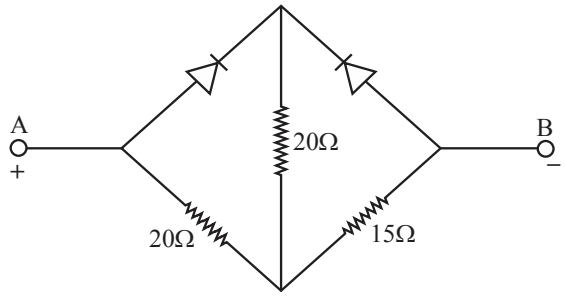
(D)

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

SC0114

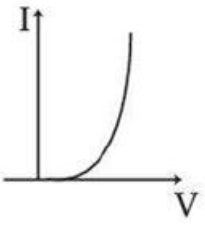
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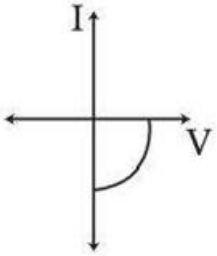
55. Two ideal diodes are connected in the network as shown in figure. The equivalent resistance between A and B is _____Ω. **[JEE Main-2022_July]**

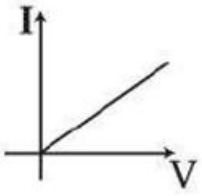


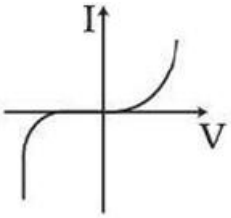
SC0115

56. Identify the solar cell characteristics from the following options : **[JEE Main-2022_July]**

(A) 

(B) 

(C) 

(D) 

SC0116

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ANSWER KEY

EXERCISE # O

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	C	C	B	C	A	D	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	C	B	B	C	A	A	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	D	C	C	C	B	C	C	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	A	D	A	C	D	C	C	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	D	D	D	C	C	A,B,D	A,C,D	B,C	B,D
Que.	51									
Ans.	B,C									

EXERCISE # S

1. 0.1 Ω

2.

	R_1	D_1	Total
V	11.35V	0.65V	12V
I	24.15 mA	24.15 mA	24.15 mA
R	470 Ω		
P	274.1 mW	15.7 mW	289.8 mW

3.

A	B	C	D	E	F	G
0	0	1	1	0	0	1
0	1	1	0	0	1	0
1	0	0	1	1	0	0
1	1	0	0	0	0	1

4. (i) 10 V (ii) 90 V (iii) 4 mA

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	1 or 3	4	4	2	2	2	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	2	4	2	2	3	4	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	3	3	2	12.00	3	2	1	1	150.00
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	25	4	2	2	100	200	25	5	5	20
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	3	3	2	4	750	15	D	B	D	D
Que.	51	52	53	54	55	56				
Ans.	C	B	14	A	25	B				

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